

Γεωδαιτική παρακολούθηση ευρύτερης περιοχής Σαντορίνης

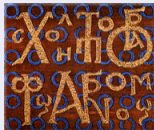
Δημήτρης Αναστασίου, Ζάνθος Παπανικολάου, Κώστας Ραπτάκης, Βαγγέλης Ζαχαρής,
Μαρία Τσακίρη

Κέντρο Δορυφόρων Διονύσου | Εργαστήριο Γεωδαισίας
Σχολή Αγρονόμων και Τοπογράφων Μηχανικών - Μηχανικών Γεωπληροφορικής
Εθνικό Μετσόβιο Πολυτεχνείο



<http://dionysos.survey.ntua.gr/>

danastasiou@mail.ntua.gr



Πέμπτη
10 Απρίλη
2025

Σαντορίνη: Επιστημονικές Διασταυρώσεις

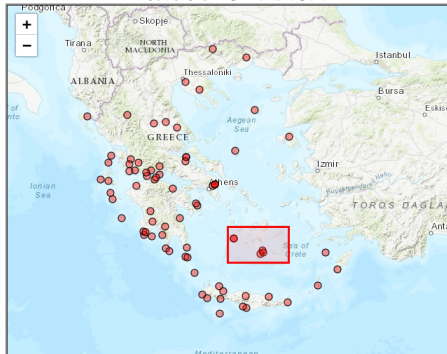
Η συμβολή των γεωδαιτικών παρατηρήσεων
και η διεπιστημονική συνεργασία
στην παρακολούθηση του ηφαιστείου της Σαντορίνης

Ενότητες παρουσίασης

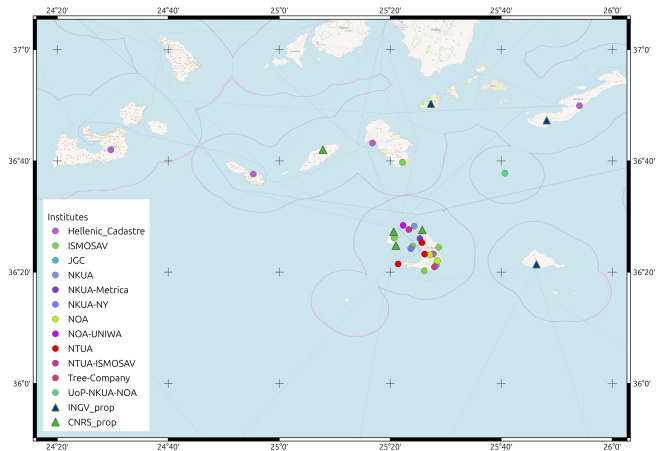
- Μόνιμοι σταθμοί GNSS
- Ανάλυση δεδομένων
- Monitoring
- Διάθεση δεδομένων/αποτελεσμάτων

Μόνιμοι σταθμοί GNSS

Δίκτυο GREECE



Δίκτυο SANTORINI



Λειτουργία σταθμών

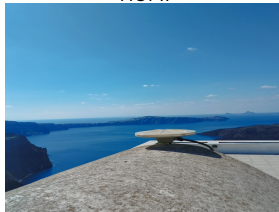
SNTR I2653M001



WNRV I2687M001



NOMI



Campaign



- Διάθεση δεδομένων: http://dionysos.survey.ntua.gr/dsoportal/_datacenter/httpdata.html
- Ο σταθμός NOMI είναι σε συνεργασία με το Πανεπιστήμιο Πατρών και την άδεια του EarthScope.

Ανάλυση δεδομένων GNSS στο ΚΔΔ

Ανάλυση δεδομένων GNSS

- Bernese v5.2 / v5.4 ([Dach et al., 2015](#))
- PRIDE-PPPAR ([Geng et al., 2019](#))
- GipsyX ([Bertiger et al., 2020](#))

Ανάλυση χρονοσειρών

- Hector v2.1 ([Bos et al., 2012](#))
- Python/C++ routines

Συχνότητα ανάλυσης

- Ημερήσιες λύσεις
 - Rapid solution - 1 day after
 - Final solution - 12 days after
- Ανάλυση υψηλών συχνοτήτων ($> 1\text{ Hz}$)
 - Δεδομένα HEPOS - 1 Hz



Ροή αυτόματης ανάλυσης

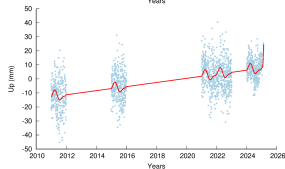
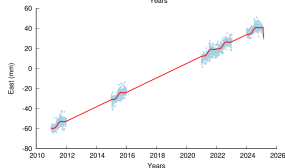
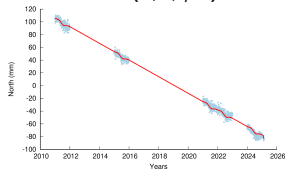


Standards:

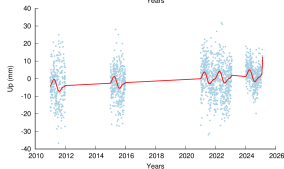
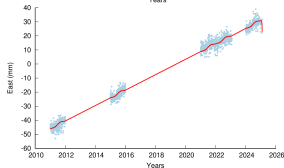
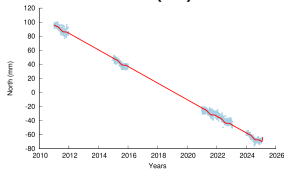
- SINEX with required info/blocks,
- Reference frame **IGS20**,
- IERS Conventions 2010,
- IGS/CODE products,
- ocean loading corrections (**FES2004**),
- 3° elevation cut-off angle; elevation dependent weighting,
- GMF and/or **VMF1**; Chen-Herring gradient parameter,
- ambiguities fixed (length-dependent algorithm),
- use GLONASS obs (when available) ☐
- use ATX files (**epn_20.atx**)

Ανάλυση χρονοσειρών

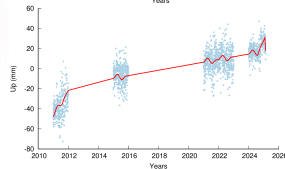
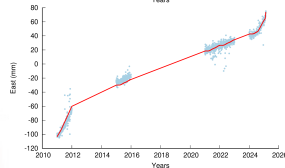
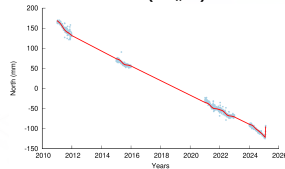
047A (Αμοργός)



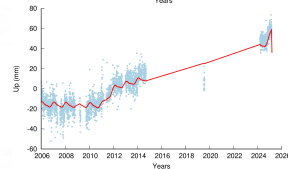
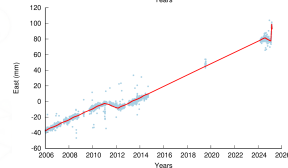
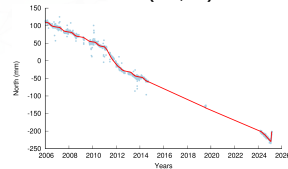
049A (Ίος)



048A (Θήρα)

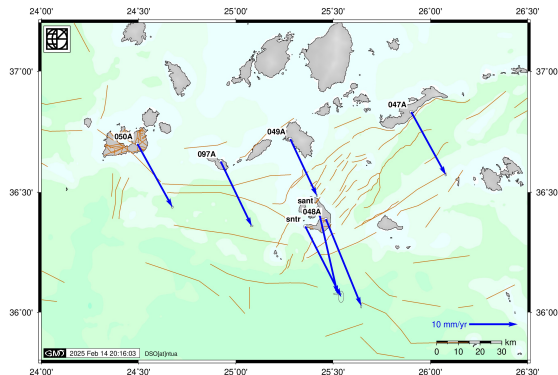


SNTR (Φάρος)

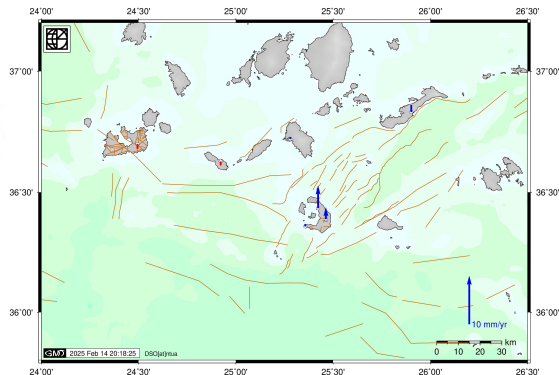


Τεκτονικές ταχύτητες [IGS20]

Horizontal Velocities 2012.3-2024.5 [IGS20]



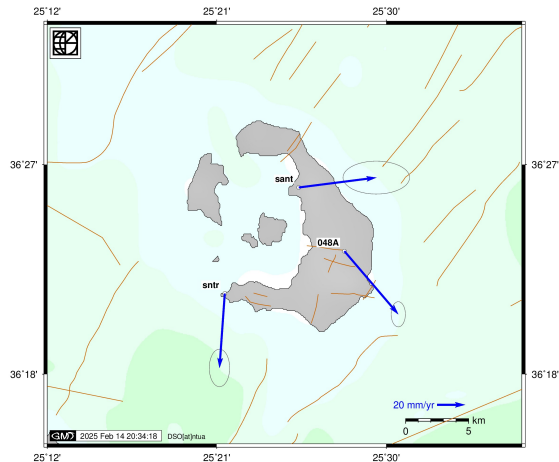
Vertical Velocities 2012.3-2024.5 [IGS20]



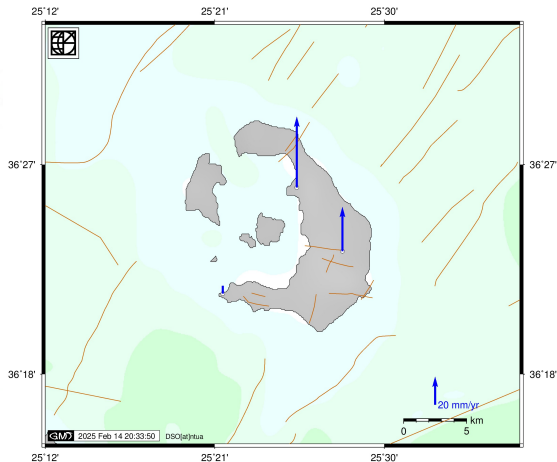
- * Δημιουργία χαρτών: Generic Mapping Tools (Wessel et al., 2019)
- * Υπόβαθρο: ETOPO1 (Amante et al., 2009)
- * Βάση ρηγμάτων: NOAfaults v6.0 (Ganas, 2024)

Μετακινήσεις 2024.6 - 2025.1

Horizontal Velocities 2024.6-2025.1 [IGS20]

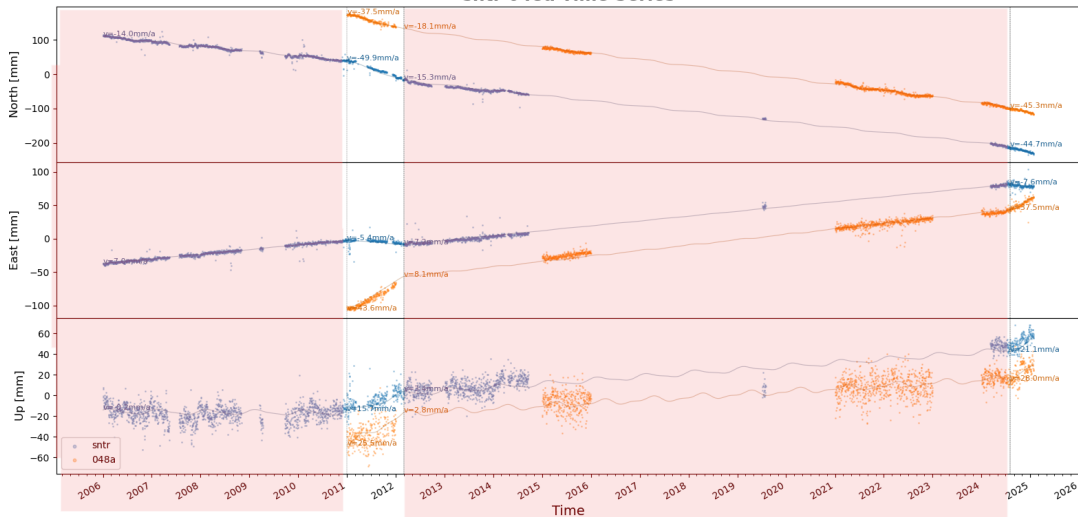


Vertical Velocities 2024.6-2025.1 [IGS20]



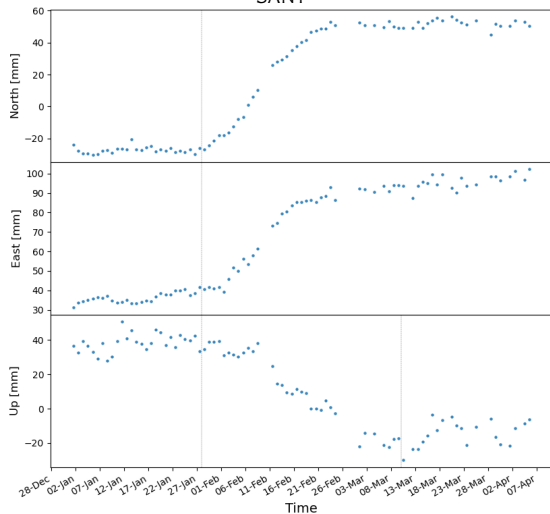
Ανάλυση χρονοσειρών

sntr 048a Time Series



Ανάλυση χρονοσειρών

SANT

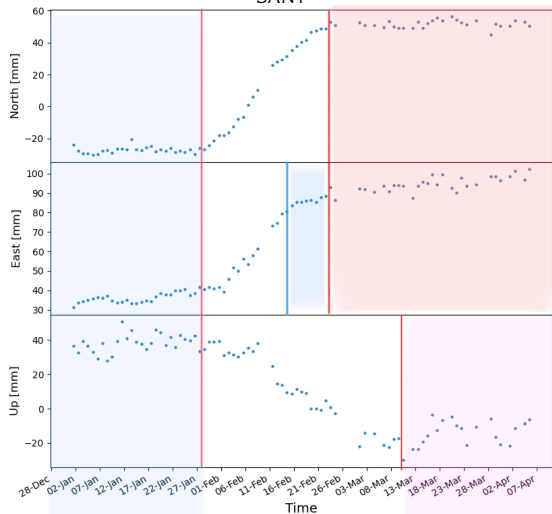


048a

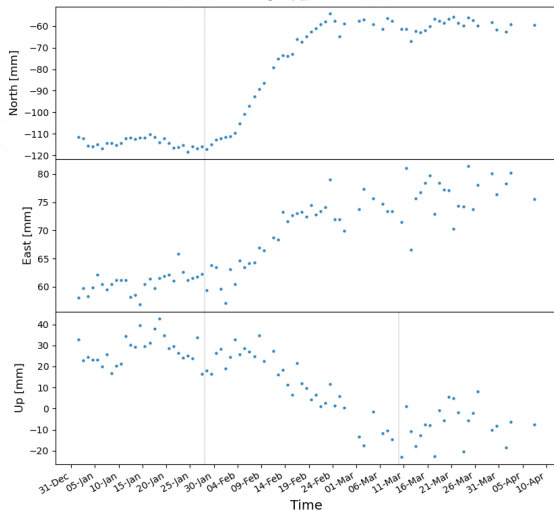


Ανάλυση χρονοσειρών

SANT

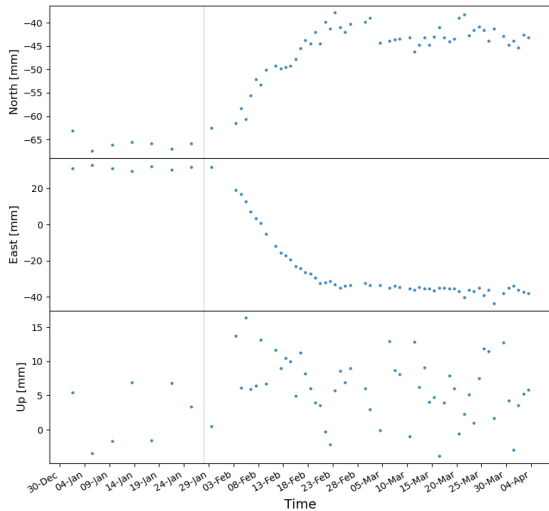


048a

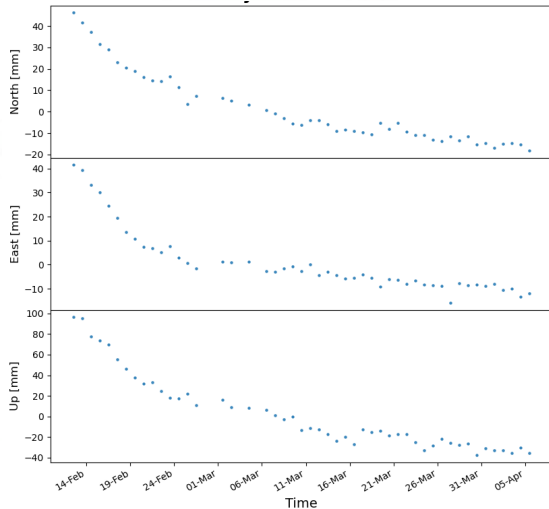


Ανάλυση χρονοσειρών

049a (Ιος)

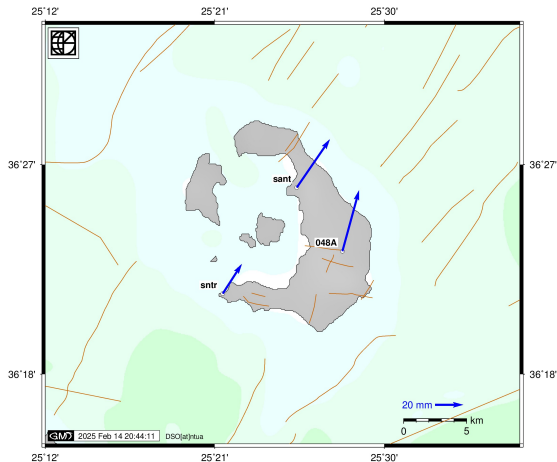


_ ANYD

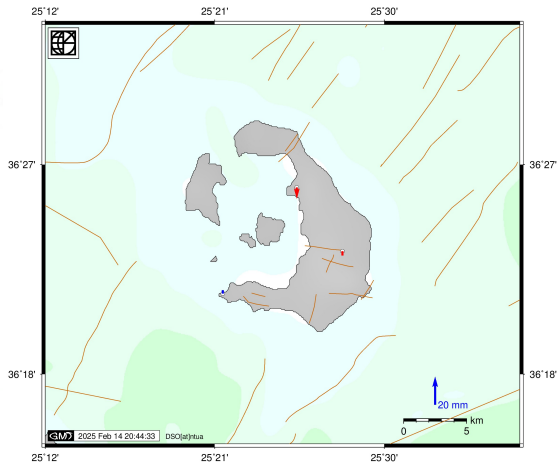


Μετακινήσεις 2025 28/1 - 16/2

Horizontal Shift 28/1-13/2 [IGS20]

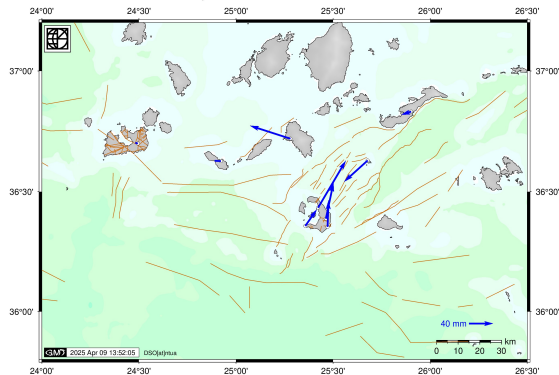


Vertical Shift 28/1-13/2 [IGS20]

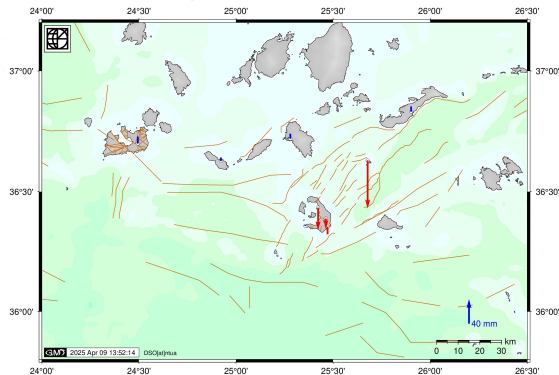


Μετακινήσεις 2025 28/1 - 25/2

Horizontal displacements 20250128-20250225 [IGS20]

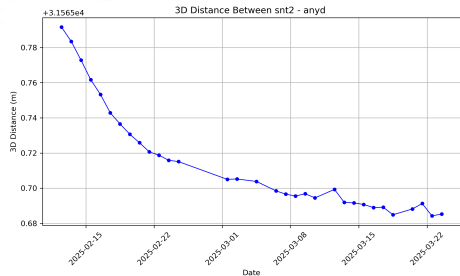
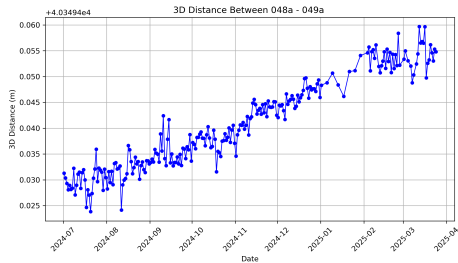
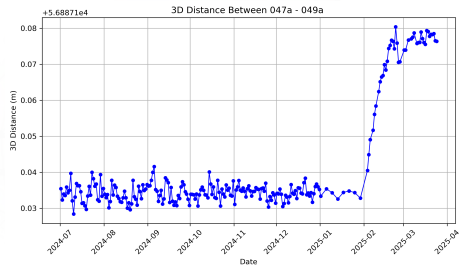
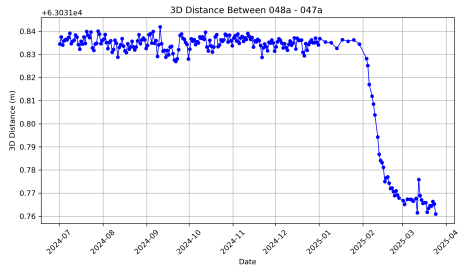


Vertical displacements 20250128-20250225 [IGS20]



Σχετικές αποστάσεις (3Δ)

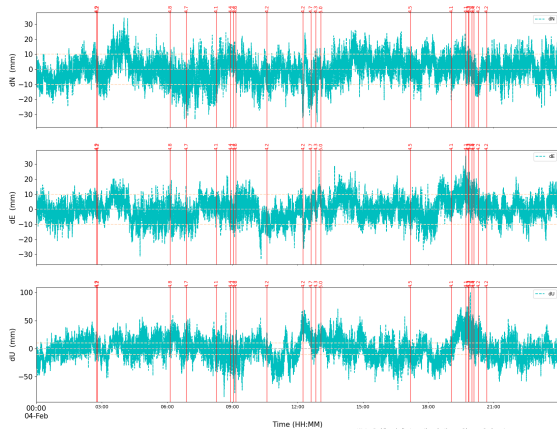
Σαντορίνη (048A) - Αμοργός (047A) - Ίος (049A) - Άνυδρος (ANYD)



Ανάλυση δεδομένων 1-s



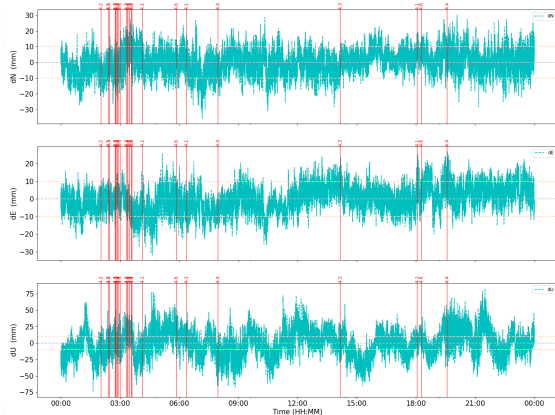
Station: 048a
Date: 2025-02-04



Note: Red lines indicate earthquake times with magnitude > 4.



Station: 048a
Date: 2025-02-06



Note: Red lines indicate earthquake times with magnitude > 4.

Διάθεση δεδομένων/αποτελεσμάτων



NATIONAL TECHNICAL UNIVERSITY OF ATHENS

DIONYSOS SATELLITE OBSERVATORY -- GNSS AREA

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Overview

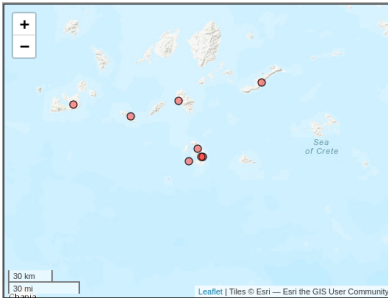
Santorini is located in the central part of the Hellenic Volcanic Arc (South Aegean Sea) and is well known for the Late Bronze Age "Minoan" eruption that might have been responsible for the decline of the great Minoan civilization on the island of Crete. Santorini has evolved over at least 650,000 years.

Collaborating Networks

- DSO-NTUA
- COMET
- Geodynamic Institute, National Observatory of Athens
- Ktimatologio SA
- University of Patra

Number of Stations

Total number of continuous stations: 8



Data and Results

Select for more information:

1. Introduction
2. History
3. Data Access
4. Data Analysis
5. Results
6. Documentation

Ευχαριστούμε για την προσοχή σας!



References I



Amante, C. and B. W. Eakins (2009). *ETOPOI 1 Arc-Minute Global Relief Model: Procedures, Data Sources and Analysis*. Tech. rep. NOAA Technical Memorandum NESDIS NGDC-24. National Geophysical Data Center, NOAA. doi: [10.7289/V5C8276M](https://doi.org/10.7289/V5C8276M). URL: <https://doi.org/10.7289/V5C8276M> (cit. on p. 8).



Bertiger, W. et al. (2020). "GipsyX/RTGx, a new tool set for space geodetic operations and research". In: *Advances in Space Research* 66.3, pp. 469–489. doi: [10.1016/j.asr.2020.04.015](https://doi.org/10.1016/j.asr.2020.04.015). URL: <https://doi.org/10.1016/j.asr.2020.04.015> (cit. on p. 5).



Bos, M. S., R. M. S. Fernandes, S. D. P. Williams, and L. Bastos (Dec. 2012). "Fast error analysis of continuous GNSS observations with missing data". In: *Journal of Geodesy* 87.4, pp. 351–360. doi: [10.1007/s00190-012-0605-0](https://doi.org/10.1007/s00190-012-0605-0) (cit. on p. 5).



Dach, Rolf, Simon Lutz, Peter Walser, and Pierre Fridez, eds. (2015). *Bernese GNSS Software Version 5.2*. University of Bern, Bern Open Publishing. doi: [10.7892/boris.72297](https://doi.org/10.7892/boris.72297). URL: <http://www.bernese.unibe.ch/docs/DOCU52.pdf> (cit. on p. 5).



Ganas, Athanassios (Aug. 2024). *NOAFAULTS KMZ layer Version 6.0*. Version version 6.0. Zenodo. doi: [10.5281/zenodo.13168947](https://doi.org/10.5281/zenodo.13168947). URL: <https://doi.org/10.5281/zenodo.13168947> (cit. on p. 8).



Geng, Jianghui, Xingyu Chen, Yuanxin Pan, Shuyin Mao, Chenghong Li, Jinning Zhou, and Kunlun Zhang (2019). "PRIDE PPP-AR: An open-source software for GPS PPP ambiguity resolution". In: *GPS Solutions* 23.91, pp. 1–10. doi: [10.1007/s10291-019-0888-1](https://doi.org/10.1007/s10291-019-0888-1). URL: <https://doi.org/10.1007/s10291-019-0888-1> (cit. on p. 5).



Wessel, P., J. F. Luis, L. Uieda, R. Scharroo, F. Wobbe, W. H. F. Smith, and D. Tian (2019). "The Generic Mapping Tools Version 6". In: *Geochemistry, Geophysics, Geosystems* 20.11, pp. 5556–5564. doi: [10.1029/2019GC008515](https://doi.org/10.1029/2019GC008515). eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2019GC008515>. URL: <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GC008515> (cit. on p. 8).